

Tutorial 1 - Revision exercise

1. Write a Helper class that allows user to
 - a. enter string, int and double from the keyboard.
 - b. print to console
2. Write a program to asks the user to enter temperature in degree Celsius and convert the temperature to kelvin and Fahrenheit.
3. Write a program that determines whether a bank customer qualifies for a loan.
To qualify, two conditions must be satisfied:
 - a. the customer must earn at least \$50,000 per year, and
 - b. the customer must have been employed at his or her current job for at least two years.
4. Write the code to print the state of the water based on the temperature of the water as given in the table below.

Temperature	State
Greater than or equal to 100 ⁰ C	Gas
Between 0 ⁰ C and 100 ⁰ C	Liquid
Less than or equal to 0 ⁰ C	Solid

5. Write a program to take in the gender into variable **gender** and amount spent into variable **amount**. The program will determine and display the free gift which a shopper will receive based on the gender and the amount spent entered as given in the table below.

Gender	Male	Male	Female	Female
Amount Spent	>= \$100	< \$100	>= \$100	< \$100
Free Gift	Shaver	None	Hand Cream	None

6. A departmental store is running a promotion event for their customers who purchase above \$200.
Different discount is given to credit cards issued by 2 different banks
 - a. Credit Cards issued by DBS – 10% discount
 - b. Credit Cards issued by OCBC – 15% discount
 - c. Credit Cards issued by other banks – 5% discount

Write a program that ask the user for total cost of purchase and banks that issued the credit card. It will display the amount that the user is to pay after

7. Write a calculator application that allows user to choose an numerical operation (add, subtract, multiple, divide, modulo and floored quotient).
The user first select the type of numerical operation and then the two operands (numbers to be operated upon). The application will then display the results of the numerical operation.
8. Write a Zoo ticketing application.
The ticketing application allows a group buy:
 - Adult-\$15
 - Child-\$10
 - Animal show is \$5 per person (group purchase)
 - Free sunshade for all person if total purchase is above \$100
 - 10% discount if spend above \$150 or is a “Friend of Zoo”
9. Declare two arrays, one with 5 person names and the other with emails for each person.
Print the name and email for each person.
10. Declare an array that can store 10 integers. Get 10 integers from the user and store them in the integer array. Sort the integers using bubble sort. (see the Bubble sort ppt)